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# MOBILE SHOP CRUD PROJECT

# Global list to store all mobile records
mobiles = []

# -----
# Function: add_mobile()
# Purpose: Add a new mobile record
# -----
def add_mobile():
    print("\n===== ADD MOBILE =====")

    mobile_id = int(input("Enter Mobile ID: "))

    # Check whether the ID already exists
    for mobile in mobiles:
        if mobile[0] == mobile_id:
            print("Mobile ID already exists.")
            return

    brand = input("Enter Brand: ")
    model = input("Enter Model: ")
    price = float(input("Enter Price: "))
    quantity = int(input("Enter Quantity: "))

    mobile = [mobile_id, brand, model, price, quantity]
    mobiles.append(mobile)

    print("Mobile added successfully.")

# -----
# Function: display_mobiles()
# Purpose: Display all mobile records
# -----
def display_mobiles():
    print("\n===== MOBILE LIST =====")

    if len(mobiles) == 0:
        print("No mobile records available.")
        return

    print("-" * 75)

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print(
    f'{"ID":<8}'
    f'{"Brand":<15}'
    f'{"Model":<20}'
    f'{"Price":<15}'
    f'{"Quantity":<10}'
)
print("-" * 75)

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for mobile in mobiles:
    print(
        f'{mobile[0]:<8}'
        f'{mobile[1]:<15}'
        f'{mobile[2]:<20}'
        f'{mobile[3]:<15.2f}'
        f'{mobile[4]:<10}'
    )

print("-" * 75)

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# -----
# Function: search_mobile()
# Purpose: Search for a mobile using Mobile ID
# -----
def search_mobile():
    print("\n===== SEARCH MOBILE =====")

    mobile_id = int(input("Enter Mobile ID to search: "))

    found = False

    for mobile in mobiles:
        if mobile[0] == mobile_id:
            print("\nMobile Found!")
            print("Mobile ID :", mobile[0])
            print("Brand   :", mobile[1])
            print("Model   :", mobile[2])
            print("Price   :", mobile[3])
            print("Quantity :", mobile[4])

            found = True
            break

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if found == False:
    print("Mobile not found.")
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# -----
# Function: update_mobile()
# Purpose: Update details of an existing mobile
# -----
def update_mobile():
    print("\n===== UPDATE MOBILE =====")

    mobile_id = int(input("Enter Mobile ID to update: "))

    for mobile in mobiles:
        if mobile[0] == mobile_id:
            print("\nMobile Found.")

            print("Current Brand   :", mobile[1])
            print("Current Model   :", mobile[2])
            print("Current Price   :", mobile[3])
            print("Current Quantity :", mobile[4])

            print("\nEnter New Details")

            new_brand = input("Enter New Brand: ")
            new_model = input("Enter New Model: ")
            new_price = float(input("Enter New Price: "))
            new_quantity = int(input("Enter New Quantity: "))

            mobile[1] = new_brand
            mobile[2] = new_model
            mobile[3] = new_price
            mobile[4] = new_quantity

            print("Mobile updated successfully.")
            return

    print("Mobile not found.")
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# -----
# Function: delete_mobile()
# Purpose: Delete a mobile record
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def delete_mobile():
    print("\n===== DELETE MOBILE =====")

    mobile_id = int(input("Enter Mobile ID to delete: "))

    for mobile in mobiles:
        if mobile[0] == mobile_id:
            print("\nMobile Found.")
            print("Brand :", mobile[1])
            print("Model :", mobile[2])

            choice = input(
                "Do you want to delete this mobile? (Y/N): "
            )

            if choice.upper() == "Y":
                mobiles.remove(mobile)
                print("Mobile deleted successfully.")
            else:
                print("Delete operation cancelled.")

        return

    print("Mobile not found.")

# -----
# Function: dashboard()
# Purpose: Display the main menu and control the application
# -----
def dashboard():

    while True:
        print("\n")
        print("=" * 45)
        print("    MOBILE SHOP MANAGEMENT")
        print("=" * 45)

        print("1. Add Mobile")
        print("2. Display All Mobiles")
        print("3. Search Mobile")
        print("4. Update Mobile")
        print("5. Delete Mobile")
        print("6. Exit")

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print("=" * 45)

choice = input("Enter your choice: ")

match choice:

    case "1":
        add_mobile()

    case "2":
        display_mobiles()

    case "3":
        search_mobile()

    case "4":
        update_mobile()

    case "5":
        delete_mobile()

    case "6":
        print("\nThank you for using Mobile Shop Management.")
        break

    case _:
        print("Invalid choice. Please try again.")
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# -----
# Function: main()
# Purpose: Starting point of the application
# -----
def main():
    dashboard()
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# -----
# Program execution starts here
# -----
if __name__ == "__main__":
    main()
```